Lie derivative

Directional derivatives of smooth functions on a manifold are defined via tangent vector acting as a derivation when applied to a real-valued function as shown in **??**. In Euclidean spaces, the directional derivative of a vector field relies on the vector space structure of $\mathbb{R}^n$ and the identification of tangent vectors with elements of $\mathbb{R}^n$. On a general manifold, this identification fails. Replacing $v \in T_pM$ with $V \in \mathfrak{X}(M)$ overcomes this, as the flow of V enables the pullback of W to p for differentiation. Hence we adopt the following definition.

Definition 1.1.1 (Lie derivative)**.** *Let M be a smooth manifold, $V \in \mathfrak{X}(M)$, and let θ be the flow of V as in Definition* **??**. *For $W \in \mathfrak{X}(M)$, the* Lie derivative of W with respect to V *is the vector field $\mathscr{L}_V W \in \mathfrak{X}(M)$ such that*

$$(\mathscr{L}_V W)_p = \frac{d}{dt}\bigg|_{t=0} d(\theta_{-t})_{\theta_t(p)}\big(W_{\theta_t(p)}\big) = \lim_{t\to 0} \frac{d(\theta_{-t})_{\theta_t(p)}\big(W_{\theta_t(p)}\big) - W_p}{t}. \tag{1.1}$$

If $\partial M \neq \emptyset$, the definition still goes through as long as V is tangent to ∂M, since this entails that the flow of V exists on all of M.

Lemma 1.1.2. *Suppose M is a smooth manifold with or without boundary, and $V, W \in \mathfrak{X}(M)$. If $\partial M \neq \emptyset$, assume in addition that V is tangent to ∂M. Then $(\mathscr{L}_V W)_p$ exists for every $p \in M$, and $\mathscr{L}_V W$ is a smooth vector field.*

The proof of this result is stated in [1, Lemma 9.36]. The next theorem is crucial to understand the role of this construction throughout the thesis.

Theorem 1.1 (Lie derivative of vector fields)**.** *If M is a smooth manifold and $V, W \in \mathfrak{X}(M)$, then $\mathscr{L}_V W = [V, W]$, the commutator between V and W.*

This theorem, proved in [1, Theorem 9.38], provides a direct geometric interpretation of the Lie bracket. It establishes that the commutator $[V, W]$ measures the failure of the flows of V and W to commute. Equivalently, it quantifies the rate at which the vector field W changes when observed from a point moving along the flow of V. This identity will be a cornerstone in the following sections, as it links the differential structure of the manifold to the algebraic properties of vector fields, and, later, to the Poisson bracket of observables.

Corollary 1.1.3. *Suppose M is a smooth manifold with or without boundary, and $V, W, X \in \mathfrak{X}(M)$.*

(a) $\mathscr{L}_V W = -\mathscr{L}_W V$.

(b) $\mathscr{L}_V [W, X] = [\mathscr{L}_V W, X] + [W, \mathscr{L}_V X]$.

(c) $\mathscr{L}_{[V,W]} X = \mathscr{L}_V \mathscr{L}_W X - \mathscr{L}_W \mathscr{L}_V X$.

(d) If $h \in C^\infty(M)$, then $\mathscr{L}_V(hW) = (Vh)W + h\mathscr{L}_V W$.

(e) If $F : M \to N$ is a diffeomorphism, then $F_*(\mathscr{L}_V X) = \mathscr{L}_{F_* V} F_* X$.

As established by [1, Corollary 9.39], these properties highlight the algebraic role of the Lie derivative. In particular, (b) and (c) show that $\mathscr{L}_V$ acts as a derivation on the Lie algebra of vector fields, while (d) confirms its behavior as a derivation with respect to multiplication by smooth functions. Property (e) establishes the naturality of the Lie derivative under diffeomorphisms, ensuring that the construction is intrinsic to the manifold structure. Now, we extend Definition 1.1.1 to differential forms.

Definition 1.1.4 (Lie derivative of a differential form)**.** *Let M be a smooth manifold, let $V \in \mathfrak{X}(M)$ have flow θ, and let $\omega \in \Omega^k(M)$ be a smooth differential form on M. The* Lie derivative of ω with respect to V *is $\mathscr{L}_V \omega \in \Omega^k(M)$ whose value at $p \in M$ is*

$$(\mathscr{L}_V \omega)_p = \frac{d}{dt}\bigg|_{t=0} \big(\theta_t^* \omega\big)_p = \lim_{t\to 0} \frac{\big(\theta_t^* \omega\big)_p - \omega_p}{t},$$

wherever this derivative exists, where $\theta_t^* \omega$ is the pullback of ω by the time-t flow map of V.

Lie differentiation satisfies a product rule with respect to wedge products.

Proposition 1.1.5. *Suppose M is a smooth manifold, $V \in \mathfrak{X}(M)$, and $\omega, \eta \in \Omega^k(M)$. Then*

$$\mathscr{L}_V(\omega \wedge \eta) = (\mathscr{L}_V \omega) \wedge \eta + \omega \wedge (\mathscr{L}_V \eta).$$

This product rule, as proved in [1, Proposition 14.33], together with the fact that the Lie derivative commutes with the exterior derivative d, makes $\mathscr{L}_V$ a derivation of the algebra of differential forms. This algebraic compatibility is essential for deriving the fundamental identity that links Lie derivatives, exterior derivatives, and interior products, commonly known as Cartan's Magic Formula.

Theorem 1.2 (Cartan's Magic Formula)**.** *On a smooth manifold M, for any smooth vector field V and any smooth differential form ω,*

$$\mathscr{L}_V \omega = \iota_V(d\omega) + d(\iota_V \omega). \tag{1.2}$$

For a complete proof of this formula, see [1, Theorem 14.35]. As we will show, it is necessary for characterizing symplectic vector fields, and it will serve as a key tool in the proof of Liouville's Theorem and the construction of the momentum map.

1.4 Hamiltonian Vector Fields and Poisson Brackets

Having built the geometric picture of phase space, this section focuses on dynamics and symmetries in Hamiltonian mechanics, where the geometric and algebraic structures introduced in the previous chapters come together. We first show how a single observable, the Hamiltonian function, representing the total energy of a physical system, generates a flow on phase space via the symplectic form. This flow governs the time evolution of all observables. The Poisson bracket, which endows the space of observables with a Lie algebra structure, encodes the compatibility between the symplectic geometry and the algebra of functions. In the following sections, this algebraic framework will be essential for the systematic analysis of symmetries and their associated conserved quantities through Noether's theorem.

Definition 1.4.1 (Hamiltonian Vector Field)**.** *Let (M, ω) be a symplectic manifold and $H : M \to \mathbb{R}$ a given C^∞-function. The vector field X_H determined by*

$$\iota_{X_H} \omega = dH, \tag{1.3}$$

called the Hamiltonian vector field *where H is called the* energy function*. (M, ω, H) is a Hamiltonian system.*

This definition selects the vector fields that arise from a global energy function H. It is useful, however, to isolate a weaker and more geometric requirement, namely that the flow of X preserves the symplectic form ω without necessarily deriving from a globally defined Hamiltonian. The next definition makes this distinction precise, and the remark that follows shows exactly how the two notions differ.

Definition 1.4.2 (Symplectic and Globally Hamiltonian Vector Fields)**.** *Let (M, ω) be a symplectic manifold as in Definition* **??***. A vector field $X \in \mathfrak{X}(M)$ is called* symplectic *or* locally Hamiltonian *if $\mathscr{L}_X \omega = 0$.*

The vector field X is called globally Hamiltonian *if there exists a smooth function $H \in C^\infty(M)$ such that the contraction satisfies $\iota_X \omega = dH$.*

Remark 1.4.3 (Closedness and exactness of Hamiltonian vector fields). The condition for a vector field X to be locally Hamiltonian, namely $\mathscr{L}_X \omega = 0$, means that it preserves the symplectic form. We can state an equivalent formulation using Cartan's magic formula as per Equation (1.2), the Lie derivative of the symplectic form is

$$\mathscr{L}_X \omega = \iota_X(d\omega) + d(\iota_X \omega).$$

Since ω is a symplectic form, it is automatically closed $d\omega = 0$. The equation reduces to:

$$\mathscr{L}_X \omega = d(\iota_X \omega).$$

Consequently the invariance of the symplectic structure stands if and only if $d(\iota_X \omega) = 0$, that is requiring the closedness of $\iota_X \omega$. Analogously, the condition for X to be globally Hamiltonian postulates the existence of H such that $\iota_X \omega = dH$. That is, by definition, requiring the exactness of $\iota_X \omega$.

The main advantage of this algebraic definition is its reliance on the symplectic structure, which maps dH into the corresponding vector field. When evaluated in local canonical coordinates, whose existence is guaranteed by Darboux's Theorem **??**, this geometric mapping recovers the classical Hamilton's equations of motion.

Theorem 1.3 (Hamilton Equations)**.** *Let $(q^1,\dots,q^n,p_1,\dots,p_n)$ be canonical coordinates for ω, so $\omega = \sum_{i=1}^n dq^i \wedge dp_i$. Then in these coordinates and dropping base points,*

$$X_H = \left(\frac{\partial H}{\partial p_i}, -\frac{\partial H}{\partial q^i}\right) = J \cdot dH, \tag{1.4}$$

here dH denotes the column vector of its coordinate partial derivatives and $J = \begin{pmatrix} 0 & \mathbb{1} \\ -\mathbb{1} & 0 \end{pmatrix}$. Thus $(q(t), p(t))$ is an integral curve of X_H if and only if Hamilton's equations hold:

$$\boxed{\dot{q}^i = \frac{\partial H}{\partial p_i}, \quad \dot{p}_i = -\frac{\partial H}{\partial q^i}, \quad i = 1,\dots,n.}$$

Proof. Let Y be defined by Equation (1.4). We then have to verify 1.3. Since by construction $\iota_Y dq^i = \partial H/\partial p_i$, $\iota_Y dp_i = -\partial H/\partial q^i$, it descends

$$\iota_Y \omega = \sum \iota_Y (dq^i \wedge dp_i) = \sum (\iota_Y dq^i) \wedge dp_i - \sum dq^i \wedge \iota_Y dp_i = \sum \frac{\partial H}{\partial p_i} dp_i + \frac{\partial H}{\partial q^i} dq^i = dH.$$

By the non-degeneracy of ω, $Y = X_H$. □

A direct physical consequence of the anti-symmetry inherent to the symplectic form is the emergence of conserved quantities which we will study in the next section. Here, we can show the main example, the Hamiltonian itself. Evaluating the time derivative of the energy along its own integral curves demonstrates that the dynamics intrinsically preserves the total energy of the system.

Proposition 1.4.4 (Conservation of energy)**.** *Let (M, ω, H) be a Hamiltonian system through Definition 1.4.1 and let $c(t)$ be an integral curve for X_H. Then $H(c(t))$ is constant in as a function of $t \in \mathbb{R}$.*

Proof. By the chain rule and Equation (1.3),

$$\begin{aligned} \frac{d}{dt} H(c(t)) &= dH(c(t)) \cdot c'(t) \\ &= \langle dH(c(t)), X_H(c(t)) \rangle \\ &= \omega\big(X_H(c(t)), X_H(c(t))\big) \\ &= 0 \end{aligned}$$

since ω is skew-symmetric. Here $\langle \cdot, \cdot \rangle$ denotes the classical dual pairing between forms and vectors. □

This conservation law suggests a more general principle, namely, any symmetry of the Hamiltonian system produces a conserved quantity. It will be formulated algebraically later that the Poisson bracket structure provides the universal mechanism for generating conserved quantities from symmetries. That is the content of Noether's Theorem 1.6.

Another important statement is that the phase flow actively preserves the underlying symplectic structure. Consequently, time evolution in classical mechanics unfolds as a continuous one-parameter group of canonical transformations, a structural feature presented by Liouville's Theorem.

Proposition 1.4.5 (Liouville's Theorem)**.** *Let (M, ω, X_H) be a Hamiltonian system, and F_t be the flow of X_H. Then for each t, $F_t^* \omega = \omega$, that is, F_t is symplectic on its domain. Thus F_t also preserves the volume form Ω_ω.*

Proof. We have

$$\begin{aligned}
\frac{d}{dt}F_t^*\omega &= \lim_{h\to 0}\frac{F_{t+h}^*\omega - F_t^*\omega}{h} \\
&= F_t^*\left(\lim_{h\to 0}\frac{F_h^*\omega - \omega}{h}\right) \\
&= F_t^*\mathscr{L}_{X_H}\omega \\
&= F_t^*\big(i_{X_H}d\omega + di_{X_H}\omega\big) \\
&= F_t^*(0 + ddH) \qquad \text{(since } d\omega = 0 \text{ and } i_{X_H}\omega = dH) \\
&= 0
\end{aligned}$$

Thus $F_t^*\omega$ is constant in t. Since $F_0 = \mathbb{1}$, it follows that $F_t^*\omega = \omega$. □

Poisson Bracket

In this section, we see that the construction above has a much deeper significance enclosed in the geometrical formulation. In fact, analysing the structure that lies beyond classical observables, we can find a Lie product given by the Poisson brackets. To show this, we start from enriching $\Omega^1(M)$ with a specific Lie bracket.

Definition 1.4.6 (Poisson Bracket of forms)**.** *Suppose (M,ω) is a symplectic manifold as in Definition* **??** *and $\alpha,\beta \in \Omega^1(M)$. The Poisson bracket is the one-form*

$$\{\alpha,\beta\} = -[\alpha^\sharp,\beta^\sharp]^\flat,$$

where $\flat$ and $\sharp$ are the musical isomorphisms as in Definition **??**.

Note that we have the following commutative diagram:

$$\begin{array}{ccc}
\mathfrak{X}\times\mathfrak{X} & \xrightarrow{-[\cdot,\cdot]} & \mathfrak{X} \\
{\scriptstyle \flat\times\flat}\downarrow & & \downarrow{\scriptstyle \flat} \\
\Omega^1\times\Omega^1 & \xrightarrow{\{\cdot,\cdot\}} & \Omega^1
\end{array}$$

As we showed in Example **??**, the commutator $[\cdot,\cdot]$ makes $\mathfrak{X}(M)$ a Lie algebra. Since $\flat$ is linear, then $\Omega^1(M)$ as a real vector space with the composition $\{\cdot,\cdot\}$ is a Lie algebra.
Our aim is to equip the observable algebra $C^\infty(M)$ with a Lie bracket. To this end, we will define a binary operation on smooth functions whose image under the exterior derivative $d : C^\infty(M) \to \Omega^1(M)$ is compatible with the Poisson bracket on forms. Establishing this compatibility will induce a Lie algebra homomorphism from $\mathfrak{X}(M)$ to the observables.

Definition 1.4.7 (Poisson Bracket of functions)**.** *Let (M,ω) be a symplectic manifold and $f,h \in C^\infty(M)$, with $X_f = (df)^\sharp \in \mathfrak{X}(M)$ as in Definition* **??***, analogously for h. The Poisson bracket between f and g is*

$$\{f,h\} = \omega(X_f,X_h) \quad \in C^\infty(M).$$

The following properties confirm that $\{\cdot,\cdot\}$ behaves as expected, namely the Poisson bracket between f and h, measures the rate of change of f along the flow of X_h. It vanishes exactly when f is conserved along such flow, giving the bracket its direct dynamical meaning.

Proposition 1.4.8. *Let (M,ω) be a symplectic manifold and $f,h \in C^\infty(M)$. Then*

$$\{f,h\} = -\mathscr{L}_{X_f}h = \mathscr{L}_{X_h}f.$$

In addition, we have that

(i) For $f_0 \in C^\infty(M)$, the map $h \mapsto \{f_0,h\}$ is a derivation.

(ii) f is constant along the orbits of $X_h \iff h$ is constant on the orbits of $X_f \iff \{f,h\} = 0$.

Proof. It holds that $\mathscr{L}_{X_f} h = i_{X_f} dh = i_{X_f} i_{X_h} \omega$ since $dh = i_{X_h}\omega$. Since $i_X i_Y \omega = -i_Y i_X \omega = \omega(Y,X)$, the last equality follows. Now, we prove the two statements below.

(i) $h \mapsto \{f_0, h\} = -\mathscr{L}_{X_{f_0}} h$ is clearly a derivation.

(ii) If F_t is the flow of X_f,

$$\frac{d}{dt}(h \circ F_t) = F_t^* \mathscr{L}_{X_f} h = -F_t^*\{f,h\},$$

which vanishes if and only if $\{f,h\} = 0$. Since $\{f,h\} = -\{h,f\}$, this is equivalent to $(d/dt)(f \circ H_t) = 0$, where H_t is the flow of X_h.

□

Observe that the identity $\{H,H\} = 0$ corresponds to conservation of energy by (ii) already proved by Proposition 1.4.4.

In this standard representation, the abstract symplectic structure reduces to its canonical form, yielding the familiar coordinate expression for the Poisson bracket.

Corollary 1.4.9. *In canonical coordinates $(q^1, \dots, q^n, p_1, \dots, p_n)$, we have*

$$\{f,g\} = \sum_{i=1}^{n} \left(\frac{\partial f}{\partial q^i} \frac{\partial g}{\partial p_i} - \frac{\partial f}{\partial p_i} \frac{\partial g}{\partial q^i} \right).$$

Remark 1.4.10. It follows from Corollary 1.4.9 that

$$\{q^i, q^j\} = 0 \qquad \{p_i, p_j\} = 0 \qquad \{q^i, p_j\} = \delta^i_j.$$

In addition to all these important properties, we can finally observe that the exterior derivative d acts as a compatible morphism, mapping the bracket of scalar observables directly to the bracket of their corresponding 1-forms on the cotangent space.

Proposition 1.4.11. *Let (M,ω) be a symplectic manifold and $f, g \in C^\infty(M)$. Then $d\{f,g\} = \{df, dg\}$.*

For a detailed proof, we refer the reader to [2, Proposition 3.3.16]. Furthermore, we can prove that the Poisson brackets are an operation suitable for making the classical observable space a Lie algebra.

Proposition 1.4.12. *The real vector space $C^\infty(M)$, together with the Poisson bracket $\{\cdot,\cdot\}$, forms a Lie algebra.*

Proof. Since d and $\omega^\sharp$ are $\mathbb{R}$ linear, the map $f \mapsto X_f$ is $\mathbb{R}$ linear. Hence $\{f,g\} = \omega(X_f, X_g)$, which entails $\{f,f\} = 0$. For Jacobi's identity, we have

$$\begin{aligned}
\{f,\{g,h\}\} &= -\mathscr{L}_{X_f}\{g,h\} = \mathscr{L}_{X_f}(\mathscr{L}_{X_g} h) \\
\{g,\{h,f\}\} &= \mathscr{L}_{X_g}(\mathscr{L}_{X_h} f) = -\mathscr{L}_{X_g}(\mathscr{L}_{X_f} h) \\
\{h,\{f,g\}\} &= \mathscr{L}_{X_{\{f,g\}}} h
\end{aligned}$$

However, $X_{\{f,g\}} = (d\{f,g\})^\sharp = \{df, dg\}^\sharp = -[(df)^\sharp, (dg)^\sharp]$ by Proposition 1.4.11. Hence $X_{\{f,g\}} = -[X_f, X_g]$ and the result follows. □

The Jacobi identity just proved a structural link between the algebra of observables and the algebra of vector fields. Specifically, the bracket of the Hamiltonian vector fields corresponding to two functions should correspond to the Hamiltonian vector field of their Poisson bracket. The following theorem formalizes this correspondence, showing that the mapping from functions to vector fields is a Lie algebra homomorphism up to a sign.

Theorem 1.4 (Lie Algebra Homomorphism of Hamiltonian Vector Fields). *Let (M,ω) be a symplectic manifold. The mapping $\mathscr{H}: C^\infty(M) \to \mathfrak{X}(M)$ defined by $f \mapsto \mathscr{H}_f = -X_f$, where $\iota_{\mathscr{H}_f}\omega = -df$, relates the Poisson algebra of smooth functions to the Lie algebra of vector fields via the identity:*

$$\mathscr{H}_{\{f,g\}} = -[\mathscr{H}_f, \mathscr{H}_g] \qquad \forall f,g \in C^\infty(M).$$

Consequently, the map $f \mapsto -X_f$ constitutes a strict Lie algebra homomorphism from $(C^\infty(M), \{\cdot,\cdot\})$ to $(\mathfrak{X}(M), [\cdot,\cdot])$.

This follows from the work of [2, Corollary 3.3.18]. It guarantees that the algebra of classical observables is intrinsically tied to the geometry of infinitesimal canonical transformations. Physically, it implies that the algebraic structure of symmetries, encoded in the Poisson bracket, is faithfully represented by the commutation of their dynamical flows on the phase space.

Corollary 1.4.13 (Lie Subalgebra of Globally Hamiltonian Vector Fields). *The space of globally Hamiltonian vector fields on a symplectic manifold (M,ω), introduced in Definition 1.4.2 as $\mathfrak{X}_{\mathscr{H}}(M)$, forms a linear subspace of $\mathfrak{X}(M)$ that is closed under the Lie bracket operation.*

For the full proof, see [2, Corollary 3.3.18].

Remark 1.4.14. The structural rigidity established in Corollary 1.4.13 follows as a direct algebraic consequence of Theorem 1.4. By definition, the space $\mathfrak{X}_{\mathscr{H}}(M)$ is the image of the map $\mathscr{H}: C^\infty(M) \to \mathfrak{X}(M)$. Since the homomorphic image of any Lie algebra necessarily forms a Lie subalgebra of the codomain, the closure of $\mathfrak{X}_{\mathscr{H}}(M)$ under the Lie bracket is automatically ensured. Such a subspace is called a *Lie subalgebra*. Furthermore, the kernel of this homomorphism consists of functions $f \in C^\infty(M)$ satisfying $df = 0$, which correspond to locally constant functions on the manifold M.

Beyond its structural role in defining symmetries, the primary physical utility of the Poisson bracket lies in its dynamical capacity to generate time evolution on the phase space.

Proposition 1.4.15. *Let X_H be a Hamiltonian vector field on a symplectic manifold (M,ω) with Hamiltonian $H \in C^\infty(M)$ and flow F_t. Then for $f \in C^\infty(M)$ we have*

$$\frac{d}{dt}(f \circ F_t) = \{f \circ F_t, H\}.$$

Proof. It follows from this calculation:

$$\begin{aligned}\frac{d}{dt}(f \circ F_t) &= \frac{d}{dt}F_t^* f = F_t^* \mathscr{L}_{X_H} f \\ &= L_{X_H}(f \circ F_t) = \{f \circ F_t, H\}.\end{aligned}$$

□

1.5 Symmetries and Geometrical Noether's Theorem

The relation between continuous symmetries and conserved quantities constitutes a key principle in theoretical physics. To analyse this correspondence formally, it is necessary to formalize the concept of physical symmetry through the Lie group actions on smooth manifolds.

In this section, we transition from the local dynamics generated by a single Hamiltonian observable to the global symmetry transformations induced by Lie groups. We shall establish how a symplectic group action on a phase space naturally generates a set of fundamental conserved observables, an algebraic and geometric synthesis culminating in the modern geometrical formulation of Noether's Theorem.

First we formalise how a Lie group can act on the points of the underlying phase space.

Definition 1.5.1 (Left Action). *Let M be a smooth manifold. A left action, which we will refer as action for simplicity, of a Lie group G on M is a smooth mapping $\Phi : G \times M \to M$ such that*

- $\Phi(e,x) = x \ \forall x \in M,$

- *given $g, h \in G$ we have $\Phi(g, \Phi(h, x)) = \Phi(gh, x)\ \forall x \in M$.*

The action $\Phi(g, \cdot)$ will be also denoted by $\Phi_g(\cdot)$.

The action of a group on the manifold identifies an orbit.

Definition 1.5.2 (Orbit)**.** *Let M be a smooth manifold and Φ be an action of G on M. For $x \in M$ the orbit is*

$$G \cdot x = \{\Phi_g(x) \mid g \in G\}.$$

Not every action explores the manifold, or the group itself, in the same way. Some collapse the whole space into a single orbit, some fail to distinguish between different group elements acting identically, and some may fix certain points while moving others freely. The next definition isolates these three qualitatively different behaviours, each of which constrains what geometric structures can later be built out of the action.

Definition 1.5.3 (Transitive, effective and free action)**.** *Let M be a smooth manifold.*

- *An action of a Lie group G on M is transitive if there is just one orbit.*
- *It is effective if $\Phi_g = \mathbb{1}$ implies $g = e$, that is $g \mapsto \Phi_g$ is one-to-one.*
- *An action is free if, for each $x \in M$, $g \mapsto \Phi_g(x)$ is one-to-one.*

While the orbit characterizes the global, geometric footprint of a group action on a manifold, its local, differential structure is governed by the continuous flow of the symmetry group. Hence, the tangent space to the orbit at any point $x \in M$ is spanned by induced velocity vector fields evaluated at x. This localized, differential perspective leads naturally to the formal definition of the infinitesimal generator.

Definition 1.5.4 (Infinitesimal generator)**.** *Let M be a smooth manifold and let G be a Lie group. Suppose $\Phi : G \times M \to M$ is a smooth action. If $\xi \in T_eG \cong \mathfrak{g}$, then*

$$\Phi^{\xi} : \mathbb{R} \times M \to M, \quad (t, x) \mapsto \Phi(\exp t\xi, x)$$

is an $\mathbb{R}$-action on M, that is, Φ^{ξ} is a flow on M. The corresponding vector field on M given by

$$\xi_M(x) = \frac{d}{dt}\Phi(\exp t\xi, x)\bigg|_{t=0}$$

is called the infinitesimal generator *of the action corresponding to ξ.*

Remark 1.5.5. The role of the exponential map in this definition serves as the bridge between abstract symmetries and physical dynamics on the manifold. Intuitively, an element $\xi \in \mathfrak{g}$ represents an infinitesimal transformation, meaning a direction of symmetry at the identity of the group. The exponential map $\exp(t\xi)$ integrates this infinitesimal tangent vector into a one-parameter subgroup, within the Lie group G, parametrized by $t \in \mathbb{R}$. By feeding this trajectory into the action Φ, we translate the abstract symmetry into a concrete flow on the physical space M. Taking the temporal derivative at $t = 0$ extracts the physical velocity field. This is the infinitesimal generator, that pushes the points of the manifold along that specific direction of symmetry.

Example 1.5.6 (Infinitesimal generator of planar rotations). Consider the canonical action of the rotation group $G = SO(2)$ on the configuration manifold $M = \mathbb{R}^2$. The Lie algebra $\mathfrak{so}(2)$ is spanned by the basis element $\xi = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$. The exponential map yields the standard rotation matrix $\exp(t\xi) = \begin{pmatrix} \cos t & -\sin t \\ \sin t & \cos t \end{pmatrix}$. Given a point $\mathbf{x} = (x, y)^T \in \mathbb{R}^2$, the infinitesimal generator associated with ξ is computed directly from the flow:

$$\xi_M(x, y) = \frac{d}{dt}\begin{pmatrix} \cos t & -\sin t \\ \sin t & \cos t \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix}\bigg|_{t=0} = \begin{pmatrix} -y \\ x \end{pmatrix}.$$

Dynamically, this yields the vector field $\xi_M = -y\frac{\partial}{\partial x} + x\frac{\partial}{\partial y}$, which is the standard generator of angular momentum and continuous rotational symmetries in the plane.

The concrete derivation of the rotational generator in the preceding example underlines a systematic correspondence. More generally, on a symplectic manifold (M,ω), the mapping $\mathfrak{g} \to \mathfrak{X}(M)$ that assigns $\xi \mapsto \xi_M$ acts as a Lie algebra anti-homomorphism, satisfying the fundamental bracket relation:

$$[\xi_M, \eta_M] = -[\xi, \eta]_M \qquad \forall \xi, \eta \in \mathfrak{g}. \tag{1.5}$$

This allows us to classify and study continuous symmetries through the local generators of the Lie algebra $\mathfrak{g}$ as shown in detail by [2, Construction on p.271]. Choosing a basis $\{e_1, \ldots, e_k\}$ of $\mathfrak{g}$ defines the structure constants $[e_\alpha, e_\beta] = c^\gamma_{\alpha\beta} e_\gamma$. The abstract commutation relations are mapped directly to the physical velocity fields on the manifold:

$$[e_{\alpha,M}, e_{\beta,M}] = -c^\gamma_{\alpha\beta} e_{\gamma,M}. \tag{1.6}$$

From this, we can infer that when the group action preserves the symplectic form, these infinitesimal generators $e_{\alpha,M}$ are necessarily locally Hamiltonian vector fields. Assuming they are globally Hamiltonian, each basis generator is constructed by a smooth scalar observable $J_\alpha \in C^\infty(M)$ such that $e_{\alpha,M} = X_{J_\alpha}$. Imposing that these Hamiltonian functions faithfully preserve the commutation relations of the symmetry group constitutes the exact geometric requirement that motivates the definition of the momentum map.

Momentum Map

We can now focus more deeply on a concept briefly outlined in the previous construction, the momentum map. This mathematical object encodes the infinitesimal generators of a group action into a single mapping, taking values in the dual of the Lie algebra. In the following, we first give the formal definition of the momentum map. Subsequently we illustrate its meaning with examples, and eventually we prove the geometric version of Noether's theorem.

Definition 1.5.7 (Momentum map)**.** *Let (P,ω) be a symplectic manifold and $\Phi : G \times P \to P$ a* symplectic action *of the connected Lie group G on P, that is, for each $g \in G$, the map $\Phi_g : P \to P$ with $x \mapsto \Phi(g,x)$ is symplectic. We say that a map*

$$J : P \to \mathfrak{g}^*,$$

where $\mathfrak{g}^*$ is the dual of the Lie algebra of G, is a momentum map *provided that, for every $\xi \in \mathfrak{g}$,*

$$d\hat{J}(\xi) = \iota_{\xi_P}\omega, \tag{1.7}$$

where $\hat{J}(\xi) : P \to \mathbb{R}$ is defined by $\hat{J}(\xi)(x) = \langle J(x), \xi \rangle$ while ξ_P is the infinitesimal generator of the action corresponding to ξ. The quadruple (P, ω, Φ, J) is called a Hamiltonian G-space.

Remark 1.5.8 (Equivalent definition of the momentum map). In other words, the request presented in Equation (1.7) can be stated as

$$X_{\hat{J}(\xi)} = \xi_P,$$

for all $\xi \in \mathfrak{g}$. This means that the fundamental vector field ξ_P associated with the Lie algebra element ξ coincides with the Hamiltonian vector field generated by the corresponding component $\hat{J}(\xi) = \langle J, \xi \rangle$ of the momentum map. Consequently, the components of the momentum map serve as the Hamiltonian functions that generate the infinitesimal action of the symmetry group on the symplectic manifold.

Before turning to concrete examples, it is worth covering briefly why the dual Lie algebra $\mathfrak{g}^*$ is the natural target space for the momentum map, rather than an arbitrary construction.

Remark 1.5.9 (Geometric Intuition of the Momentum Map). The abstract codomain of J, namely the dual of the Lie algebra $\mathfrak{g}^*$, can initially appear counterintuitive. In classical mechanics, an observable is inherently a real-valued function on the phase space. A continuous symmetry is generated by an element of the Lie algebra, $\xi \in \mathfrak{g}$, per Definition 1.5.4. To systematically assign a scalar conserved quantity to each possible symmetry direction ξ, we require a map that linearly takes a vector in $\mathfrak{g}$ and yields a real number at each point $x \in P$. The mathematical space is exactly the dual space $\mathfrak{g}^*$. Therefore, evaluating $J(x) \in \mathfrak{g}^*$ on a generator $\xi \in \mathfrak{g}$ through the natural pairing $\hat{J}(\xi)(x) = \langle J(x), \xi \rangle$ yields the observable $\hat{J}(\xi)(x)$. Geometrically, this translates to taking the scalar product between an element representing the generalized momentum and a tangent vector representing the infinitesimal symmetry transformation.

With this interpretation in hand, we make the abstract construction concrete in the simplest possible setting, a free particle.

Example 1.5.10 (Translational Symmetry and Linear Momentum). Consider the phase space of a free particle, $P = T^*\mathbb{R}^3 \cong \mathbb{R}^3 \times \mathbb{R}^3$, with canonical coordinates (q, p) and symplectic form $\omega = \sum dq^i \wedge dp_i$. Let $G = \mathbb{R}^3$ act on P via spatial translations: $\Phi_a(q, p) = (q + a, p)$. The Lie algebra $\mathfrak{g}$ is isomorphic to $\mathbb{R}^3$, and its elements ξ represent constant translation vectors.

The infinitesimal generator on P is the constant vector field $\xi_P = \sum \xi^i \frac{\partial}{\partial q^i}$. By solving the defining relation $d\hat{J}(\xi) = \iota_{\xi_P}\omega$, one finds $\hat{J}(\xi)(q, p) = \langle p, \xi \rangle$. Consequently, the momentum map $J : P \to (\mathbb{R}^3)^*$ is $J(q, p) = p$, which is the standard classical linear momentum.

The translational example above relied on the global exactness of the symplectic form, a convenient generic circumstance. It is natural to ask what happens when this hypothesis fails, or more generally when the infinitesimal generators of the action are only locally, rather than globally, Hamiltonian.

Remark 1.5.11 (Limitations for momentum map). If the infinitesimal generators ξ_P associated with the action are only locally Hamiltonian, we cannot define a global momentum map. According to Definition 1.5.7, the momentum map requires the existence of a family of global smooth functions $\hat{J}(\xi)$ on P whose differential satisfies $d\hat{J}(\xi) = \iota_{\xi_P}\omega$. If the 1-form $\iota_{\xi_P}\omega$ is closed but not exact, such global functions do not exist on the manifold. This prevents the formulation of the momentum map.

The following theorem, proved in [2, Theorem 4.2.10], establishes a systematic procedure to compute the momentum map directly, relying upon the critical hypothesis that the symplectic manifold admits an exact symplectic form whose globally defined primitive is strictly invariant under the designated group action.

Theorem 1.5. *Let Φ be a symplectic action of a connected Lie group G on a symplectic manifold (P, ω). Assume ω on P is exact, $\omega = -d\theta$ and the action leaves θ invariant. That is, $\Phi_g^*\theta = \theta$ for all $g \in G$. Then $J : P \to \mathfrak{g}^*$ defined by*

$$\langle J(x), \xi \rangle = (\iota_{\xi_P}\theta)(x),$$

is a momentum map for that action.

Example 1.5.12 (Momentum map for translational symmetry via exact forms). The preceding theorem provides an algorithmic method to compute the momentum map in the case of cotangent spaces $P = T^*Q$, where the canonical symplectic form is structurally exact. We consider again the phase space of a free particle $P = T^*\mathbb{R}^3 \cong \mathbb{R}^3 \times \mathbb{R}^3$, equipped with canonical coordinates (q, p). By defining the canonical 1-form as $\theta = \sum_{i=1}^3 p_i dq^i$, its exterior derivative is given by:

$$d\theta = \sum_{i=1}^{3} dp_i \wedge dq^i = -\sum_{i=1}^{3} dq^i \wedge dp_i = -\omega. \tag{1.8}$$

Thus, we recover the required condition $\omega = -d\theta$.
The action of the translation group $G = \mathbb{R}^3$ operates on the phase space via $\Phi_a(q, p) = (q + a, p)$. We first verify that this action preserves the 1-form θ:

$$\Phi_a^*\theta = \sum_{i=1}^{3} (p_i \circ \Phi_a) d(q^i \circ \Phi_a) = \sum_{i=1}^{3} p_i d(q^i + a^i) = \sum_{i=1}^{3} p_i dq^i = \theta. \tag{1.9}$$

Since θ is strictly invariant under the global action, the hypotheses of the theorem are fully satisfied. The infinitesimal generator associated with the Lie algebra element $\xi \in \mathbb{R}^3$ is the constant vector field $\xi_P = \sum_{i=1}^3 \xi^i \frac{\partial}{\partial q^i}$. Evaluating the contraction of θ along ξ_P directly yields the momentum observable:

$$\langle J(q, p), \xi \rangle = (\iota_{\xi_P}\theta)(q, p) = \left(\sum_{i=1}^{3} p_i dq^i \right) \left(\sum_{j=1}^{3} \xi^j \frac{\partial}{\partial q^j} \right) = \sum_{i=1}^{3} p_i \xi^i = \langle p, \xi \rangle. \tag{1.10}$$

This algebraic procedure cleanly and instantaneously yields the linear momentum map $J(q, p) = p$.

Noether's Theorem

The momentum map is the structure that assigns to each infinitesimal symmetry generator a scalar observable whose Hamiltonian vector field coincides with that generator. In other words, the momentum map lifts the abstract Lie algebra of symmetries into the concrete Poisson algebra of observables on phase space. This lift is precisely what allows us to translate invariance of the Hamiltonian under the group action into conservation laws. The modern geometric formulation of Noether's Theorem, presented below, establishes this correspondence rigorously and shows that the momentum map is the central unifying object linking symmetries, conserved quantities, and the algebraic structure of classical mechanics.

Theorem 1.6 (Geometric Noether Theorem). *Let $\Phi : G \times P \to P$ be a symplectic action of a connected Lie group G on a symplectic manifold (P, ω) admitting a momentum map $J : P \to \mathfrak{g}^*$ according to Definition 1.5.7. A Hamiltonian function $H : P \to \mathbb{R}$ is invariant under the action of G, that is*

$$H(\Phi_g(x)) = H(x) \quad \forall x \in P,\ \forall g \in G,$$

if and only if the momentum map J is an integral of the motion for the Hamiltonian vector field X_H such that

$$J(F_t(x)) = J(x) \quad \forall x \in P,\ \forall t \in \mathbb{R},$$

where F_t denotes the flow of X_H.

Proof. Both invariance statements are equivalent to the pointwise vanishing of the bracket $\{H, \hat{J}(\xi)\}$ for every $\xi \in \mathfrak{g}$, where $\hat{J}(\xi) := \langle J(\cdot), \xi \rangle$. We show the chain of equivalences.
Since G is connected, it is generated by $\exp(\mathfrak{g})$: Every $g \in G$ is a finite product $\exp(\xi_1) \cdots \exp(\xi_k)$. Hence

$$H(\Phi_g(x)) = H(x)\ \ \forall g \in G \quad \Longleftrightarrow \quad H(\Phi_{\exp(t\xi)}(x)) = H(x)\ \ \forall \xi \in \mathfrak{g},\ \forall t \in \mathbb{R},$$

the backward implication holding because invariance under each individual flow $(\xi_i)_P$ composes to invariance under the product g. Differentiating the expression on the RHS in t at $t = 0$ yields

$$\langle dH(x), \xi_P(x) \rangle = 0 \quad \Longrightarrow \quad \mathscr{L}_{\xi_P} H = 0 \quad \forall \xi \in \mathfrak{g}.$$

By the equivalent definition of the momentum map in Remark 1.5.8, $\xi_P = X_{\hat{J}(\xi)}$, hence

$$\mathscr{L}_{\xi_P} H = \mathscr{L}_{X_{\hat{J}(\xi)}} H = \{H, \hat{J}(\xi)\}.$$

By skew-symmetry of the Poisson bracket it descends that

$$\{H, \hat{J}(\xi)\} = 0 \quad \Longleftrightarrow \quad \{\hat{J}(\xi), H\} = \mathscr{L}_{X_H} \hat{J}(\xi) = 0.$$

At last, $\mathscr{L}_{X_H} \hat{J}(\xi) = 0$ for all t is equivalent to $\hat{J}(\xi)$ being constant along the flow F_t of X_H,

$$\hat{J}(\xi)(F_t(x)) = \hat{J}(\xi)(x) \quad \forall t \in \mathbb{R},\ \forall \xi \in \mathfrak{g},$$

and, since the pairing between $\mathfrak{g}^*$ and $\mathfrak{g}$ is non-degenerate, this holds for all $\xi \in \mathfrak{g}$ if and only if

$$J(F_t(x)) = J(x) \quad \forall t \in \mathbb{R}.$$

Chaining all the equivalences together proves the theorem. □

The proof of the geometric Noether theorem presented above reveals that the entire argument operates at the infinitesimal level. It uses only the Lie algebra $\mathfrak{g}$ and its action on phase space via the infinitesimal generators. Consequently, the conservation of the momentum map follows purely from the algebraic compatibility between the Poisson bracket and the Lie bracket of vector fields, as established in Theorem 1.4. This underscores that the theorem is fundamentally a statement about Lie algebras and their representation on the Poisson algebra of observables.

Remark 1.5.13 (Conservation along the Flow). The deeper geometrical meaning of this theorem can be seen from in its proof. The derivation demonstrates that the directional derivative of H along the symmetry generator ξ_P vanishes, $\mathscr{L}_{\xi_P} H = 0$. In turn, it holds that $\mathscr{L}_{X_H} \hat{J}(\xi) = 0$. Thus, the observable $\hat{J}(\xi)$ remains strictly constant when transported along the dynamical orbits generated by X_H.